Phishing Email Detection & Awareness Report

Prepared by: Mimansha Mishra

Internship: Cyber Security – Future Interns

Task: Task 2 – Phishing Detection & Awareness System

Date: 19-04-2026

1. Introduction

Phishing is one of the most common cyber attacks used today. In this type of attack, fake emails are sent to users to trick them into sharing sensitive information like passwords, OTPs, or bank details.

These emails often look real and create a sense of urgency or fear so that users act quickly without thinking. Sometimes they even offer rewards or opportunities to make the email more convincing.

In most cases, the system is not weak—the user is simply unaware. That is why understanding phishing emails is very important.

In this project, I analysed multiple phishing email samples, identified common warning signs, and explained how such attacks work. The aim is to help users recognize and avoid these threats.

2. Methodology

To complete this task, I used sample phishing emails and analysed them step by step.

For each email, I checked:

- Who the sender appears to be
- The tone and message of the email
- Any links included in the email
- Common phishing warning signs

I did not click on any links. All analysis was done safely by just inspecting the content.

Each email was then classified as Safe, Suspicious, or Phishing based on the findings.

3. Email Analysis

Email 1: Account Lock Warning

Subject: Urgent: Your Account Will Be Locked

This email tries to scare the user by saying their account will be locked if they don't act immediately.

Key Observations:

- The sender is not clearly mentioned
- Uses urgent and threatening language
- Greeting is very generic ("Dear Customer")
- A suspicious link is provided

Risk Level: Phishing

Explanation:
The email creates panic so the user clicks the link without thinking. The link likely leads to a fake website that looks real. Once the user enters their details, the attacker can steal their credentials.

Possible Impact:

- Account hacking
- Financial loss
- Personal data theft

Email 2: Job Offer Scam

Subject: Congratulations! You Are Selected

This email claims that the user has been selected for a job and asks for payment and personal details.

Key Observations:

- No proper company name or details
- Asks for a processing fee (which is unusual)
- Suspicious link with an unknown domain

- Message is vague and not professional

Risk Level: Phishing

Explanation:
This is a common scam targeting job seekers. The attacker tries to gain trust by offering a job, then asks for money or personal details.

Possible Impact:

- Loss of money
- Exposure of personal information
- Risk of identity misuse

Email 3: Delivery Failure Scam

Subject: Delivery Failed – Action Required

This email says a package delivery failed and asks the user to update details.

Key Observations:

- No mention of a real delivery company
- Generic greeting ("Dear User")
- Suspicious link
- Creates urgency to act quickly

Risk Level: Phishing

Explanation:
This email uses a common situation (parcel delivery) to trick users. The link may ask for personal or payment details.

Possible Impact:

- Theft of personal data
- Financial fraud
- Further scam attempts

Email 4: Password Reset Alert

Subject: Password Reset Request

This email says a password reset was requested and asks the user to take action.

Key Observations:

- Sender is not verified
- No official branding
- Suspicious reset link
- Creates fear of account compromise

Risk Level: Phishing

Explanation:
This is a credential harvesting attack. The user is tricked into entering login details on a fake reset page.

Possible Impact:

- Account takeover
- Loss of sensitive information
- Unauthorized access

4. Summary of Findings

Email Type	Risk Level	Main Issue
Account Lock Warning	Phishing	Urgency + Fake Link
Job Offer Scam	Phishing	Payment + Fake Opportunity
Delivery Failure Scam	Phishing	Fake Scenario + Data Theft
Password Reset Scam	Phishing	Credential Harvesting

5. Awareness and Prevention

Phishing attacks can be avoided if users stay careful and follow some basic practices.

Do's:

- Always check the sender's email address
- Look closely at links before clicking
- Visit official websites directly instead of using email links

- Enable two-factor authentication (2FA)
- Report suspicious emails

Don’ts:

- Don’t click on unknown links
- Don’t share passwords or OTPs
- Don’t trust urgent messages blindly
- Don’t download attachments from unknown sources

6. Conclusion

Phishing attacks are simple but very effective because they target human behaviour. Many users fall for these attacks because they are not aware of how phishing works.

Through this project, I learned how to identify phishing emails and understand the tricks used by attackers. Small signs like a suspicious link or urgent message can help detect these attacks early.

Creating awareness is one of the best ways to prevent phishing. If users stay alert and follow basic precautions, most of these attacks can be avoided.